



Combining Interpretable Embedded Multicriteria Feature Cross-Selection Engineering and Machine Learning to Mimic the Brain for Stock Trading Signal Prediction

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Abstract

Stock trading signal prediction is very important for investors' trading decisions. However, since the stock market is a complex and nonlinear system, stock trading is frequent and complex. Human beings cannot integrate all the relevant information in time and make the right decisions by their brains alone. Machine learning can mimic the brain, learn from experience, and discover the connection between different things, thus realizing correct prediction and decision-making. Therefore, this study innovatively proposes a fusion of interpretable embedded multicriteria feature cross-selection engineering to capture effective features. Meanwhile, an optimized neural network prediction model is proposed where the Bayesian (BO) algorithm assumes the task of searching for hyperparameter combinations. The methods are as follow: (1) Daily stock prices are categorized into four types of key points for stock trading signals based on the time series extreme point algorithm. (2) A more comprehensive range of impact factors is constructed. Starting from the stock's historical trading data, based on the stock's trend, volatility, and turnover flow, five categories of technical indicators are constructed: Overlap Study, Momentum Indicator, Momentum Indicator, Volatility Indicator, and Price Conversion. (3) To construct a feature cross-selection method with multiple feature screening criteria to find the optimal feature influencing factors from different evaluation dimensions. (4) The hyper-parameters of the Artificial Neural Network (ANN) are optimized using Bayesian optimization algorithm. The optimized ANN is then used to model the data and obtain predictions. Twenty stocks were randomly selected from Shanghai Stock Exchange and Shenzhen Stock Exchange as experimental data to verify the validity of the model. The accuracy of the model proposed in this paper is 54.83%, 55.46%, and 54.70% for stocks with upward, steady, and downward trends respectively. The accuracy is on average 7.93%, 8.09%, and 8.09% higher than the comparison model. The return on investment through the predicted results of the model is 21.87%, 7.76%, and -3.51% respectively, which is better than the other comparative models. It can be seen from the experiments that the feature cross-selection method with multi-feature screening criterion can help the model to better find the optimal feature influencing factor, which helps to improve the accuracy of prediction. The Bayesian optimization algorithm contributes to the performance improvement of the ANN. After modeling the features using the Bayesian optimized ANN, the stock trading signal prediction model proposed in this paper is significantly better than other prediction models.

Keywords Multicriteria feature cross-selection engineering · Bayesian optimization · Machine learning · Stock trading signal prediction

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Introduction

The stock market can reflect the economic situation of a country and occupies a crucial position in a country's economic system [1]. In addition, the stock market is not only an important financing channel for companies, but also an important investment channel for common investors. Therefore, it is significant to realize the accurate prediction of stocks. However, the stock market has very many

uncertainty risks [2], The stock system is a complex system with instability [3] and non-linearity [4, 5], and the stock data are highly volatile [6], non-stationary [7], and noisy [8], so it is not easy to achieve accurate stock prediction. Therefore, stock forecasting has been a research topic of great interest to researchers, and it is challenging to construct a stock forecasting model with excellent forecasting performance.

Currently, research methods for stock forecasting are divided into four main categories, fundamental analysis [9], technical analysis, statistical methods [10], and machine learning methods [11]. Fundamental analysis is used to make trading decisions through information on macroeconomic conditions, industry dynamics, and other information. Due to factors such as the expected individual inconsistency of the country's macroeconomic situation [12], making stock trading decisions through fundamental analysis alone is highly subjective and restrictive. Technical analysis assumes that historical stock prices and technical indicators will follow a fixed trend, and that future stock prices will fluctuate according to this fixed trend. However, because of the chaotic, noisy, and nonlinear characteristics of financial time series data [13], its trends usually do not follow a fixed pattern. Therefore, fundamental and technical analyses are not good predictors of stock markets. The main idea of statistical analysis methods is to use statistical models, such as autoregressive integrated moving average (ARIMA) [14, 15], generalized autoregressive conditional heteroskedasticity (GARCH) [16], and others, to forecast stocks. However, statistical models often struggle to achieve desirable results in financial time series forecasting due to the strict requirements of statistical models for the smoothness of the series and their limitations in dealing with nonlinearly related data.

In recent years, machine learning has gained popularity among researchers due to its ability to handle stochastic, chaotic, and nonlinear data in the stock market [17]. Machine learning technology is the current frontier technology in the field of computing, which has a strong ability to capture and analyze data [18]. Currently, machine learning has been increasingly and deeply applied in stock prediction research such as tree models [19], support vector machines (SVM) [20, 21], artificial neural networks (ANN) [22, 23], and convolutional neural networks (CNN) [24] and so on. All these machine learning models are capable of handling high-dimensional nonlinear data. Lahmiri [25] applied the SVM model to intra-day high noise stock prices and compared it with four other techniques widely used for stock market price prediction, and their results showed that using SVM for predicting stock prices can achieve satisfactory results. Although the SVM model can solve high-dimensional and nonlinear problems and improve model prediction performance [26], it is more sensitive to missing values and requires careful selection of the kernel function [27]. If the

kernel function is not selected properly, it can significantly reduce the prediction performance.

The ANN model has a strong parallel processing capability, is robust to noisy data, can adequately approximate complex nonlinear relationships, and has a high classification accuracy [28]. Guresen et al. [29] argue that ANN is one of the best techniques for stock market modeling because ANN models do not rely on standardized formulas and can be easily adapted to market changes. Schwartz and Whitcomb [30] used IBM daily prices and used ANN to forecast the stock market, and they found that ANN were able to identify nonlinear relationships in the series, as well as unknown patterns in stock price changes. Yakup Kara et al. [31] selected ten representative technical indicators for stock prediction on ANN and SVM models, respectively, and they found that the ANN model has significantly better predictive performance than the SVM model.

The ANN models can yield more desirable results in the field of stock prediction. However, there are some existing stock prediction studies that deserve improvement. First, some stock forecasting researches use a single dimensional feature, the daily closing price of a stock, as an input to the model, without taking into account the effect of relevant features on stock prices. Some scholars have considered this problem and tried to input historical stock trading data together with technical indicators into the forecasting model as the input features of the model, but they usually input all features directly into the forecasting model, ignoring the importance of feature selection [32]. Feature selection achieves the goal of improving model accuracy at the same time reducing computational effort by eliminating irrelevant and redundant features. Therefore, feature selection is particularly important [33]. As a result, some researchers have started to incorporate feature screening methods into stock prediction models to remove redundant features to improve the generalization performance and efficiency of the models. However, many studies have selected only one method of dimensionality reduction. Different dimensionality reduction methods differ in the feature selection criteria, and using only a single feature filtering method will cause local deviations in the selected features [34], those that meet this dimensionality reduction criterion will be left behind, while those that do not will be deleted, and the feature set will be shifted toward this dimensionality reduction criterion, and the selected features will not adequately represent the data information, thus reducing the performance of the model. K Best [35] is one of the most basic methods in feature filtering, which uses statistical related knowledge to evaluate the importance of features, and filters the features by determining which features are more important through the high or low scores. LightGBM [36] can learn the features that are best for improving the accuracy of the model given the model. It is simply the process of determining the model

and selecting those features that are important for the training of the model. Liu et al. [37] combined Shapley additive interpretation with four classifiers for the diagnosis of Parkinson's disease. The experimental results showed that the four models based on SHAP value feature selection achieved better classification results compared with F scores, analysis of variance, and mutual information (MI) feature selection methods. Combining machine learning models and SHAP values for feature selection can not only evaluate the importance of each feature, but also have some interpretability.

Secondly, there are still some areas for improvement in the current ANN algorithm. (1) Traditional ANN models tend to fall into overfitting and get trapped in local minima [38]. (2) One way to achieve more effective machine learning stock prediction models is to use new and improved algorithms. ANN models contain many hyperparameters, and to obtain a superior stock prediction model must choose the appropriate hyperparameters. Wang and Zhu [39] used a genetic algorithm (GA) to optimize several parameters of an ANN and applied it in the closing price prediction of stock indices, and their empirical results showed that a machine learning prediction model with intelligent optimization could obtain satisfactory prediction results. However, most of the previous intelligent optimization algorithms do not sufficiently take into account the previous evaluation results when determining the hyperparameter. The Bayesian optimization method [40] adopts a Gaussian process that takes into account the evaluation results of previous parameters and continuously updates the prior function with few parameter iterations, which is faster. In addition, Bayesian optimization mitigates the problem that ANN models tend to be trapped in local minima, and it is still more robust to non-convex problems and can obtain globally optimal solutions.

Lastly, the above literature suggests that most current stock forecasting models predict short-term stock price changes. However, the stock prices predicted by these models only tell investors short-term information about stock price movements, which are constantly changing on a daily basis. It is impossible to know when the best time to trade a stock is by the price movements alone. To make a stock buying or selling decision, investors need to combine the predicted stock price with their own experience to analyze whether a certain price movement is a signal for stock trading. Therefore, in real financial market trading, the average investor is primarily concerned with stock trading signals, which means making stock buying and selling decisions at the right time rather than just getting the mere price movement of the stock.

Based on this, a novel combination of embedded multicriteria cross-feature selection engineering with interpretability and Bayesian-optimized neural network algorithm for stock trading signal prediction (IEMFCE-BO-ANN) is proposed. First, construct the stock trading

signal prediction problem as a classification problem. Then, the initial feature pool is constructed, and all feature factors associated with the variables are included in the initial feature pool. In order to make the data features fully represent the implied information in the data with the least number of features, three different dimensionality reduction techniques are applied to the initial feature pool to calculate the importance degree of each feature separately, and then the filtered features under different criteria are cross-selected to generate the final feature set, where each feature in the feature set can satisfy the multiple dimensionality reduction criteria, and can represent the implicit information in the data comprehensively. Finally, the feature-selected data are used as model inputs to predict the stock trading signals using a Bayesian-optimized ANN model. To verify the performance of the proposed model, five other models were selected for comparative analysis, and all models were compared in terms of both prediction accuracy and investment return.

The main contributions of this study are as follows.

- i. More integrated and comprehensive stock impact factors are mined to improve model prediction accuracy. From the historical trading data of the stock, five categories of technical indicators, Overlap Studies, Momentum Indicators, Volume Indicators, Volatility Indicators, and Price Transform, were constructed from five aspects.
- ii. An innovative feature cross selection method with multiple feature screening criteria is proposed. The three approaches are LightGBM, K Best, and SHAP values based on the XGBoost algorithm. Among them, LightGBM conducts feature selection from the perspective of model performance. From the perspective of model interpretability, XGBoost is combined with game-theoretic SHAP values to represent the importance of each feature using the absolute value of the SHAP averaged over the overall sample of features. K Best is a filtered feature selection algorithm, which performs feature optimization from a statistical perspective.
- iii. An optimized neural network prediction model is proposed where the Bayesian (BO) algorithm assumes the task of searching for hyperparameter combinations. This intelligent optimization algorithm, without any qualification of the objective function, simulates the objective function by sampling it until the global optimum is found. In addition, the Bayesian optimization algorithm is able to continuously update the replacement function by considering the results of previous parameter information evaluation when determining the hyperparameter values, with fewer and faster parameter iterations.

- iv. Currently, there are fewer studies on stock trading turning points compared to stock price prediction. However, for the ordinary investor, the main concern in trading in the financial markets is the buying and selling decisions rather than the daily price. Therefore, the paper can enrich this area and provide investors with more comprehensive investment advice.

The rest sections of this paper are organized as follows. Section 2 introduces three dimensionality reduction techniques, ANN models, and Bayesian optimization techniques. Section 3 details the framework of the IEMFCE-BO-ANN model. Section 4 conducts an empirical analysis and validates the performance of the mode. Finally, Sect. 5 summarizes the work of this paper and provides some brief observations.

Methodology

Light Gradient Boosting Machine (LightGBM)

The idea of LightGBM algorithm is to build multiple decision trees on the basis of massive data and integrate the output of multiple decision trees into the final output of the model. A weak learner $k_t(x)$ is obtained for each iteration of training and the goal is to minimize the final loss function $L(y, k_t(x))$. The detailed process of LightGBM can be found in [36].

$$L(y, k_t(x)) = L(y, k_{t-1}(x) + k_t(x)) \quad (1)$$

Where $k_{t-1}(x)$ represents the last iteration of the strong learner, and $L(y, k_{t-1}(x))$ represents the loss function from last iteration.

Extreme Gradient Boosting (XGBoost)

XGBoost is an improvement to the GBDT algorithm, which is also a decision tree algorithm. XGBoost integrates multiple weak learners into one strong learner and realizes the common decision of multiple decision trees. XGBoost is composed of multiple classification and regression trees Classification and Regression Tree (CART), so the integration of multiple decision tree results makes the final result more accurate. Its algorithm model is as follows:

$$y_i = \sum_{l=1}^L f_l(x_i) f_i \in F \quad (2)$$

where, y_i denote the prediction value for i^{th} data, F represent the ensemble space of tree models, and x_i represents the feature vector for i^{th} data. The detailed process of XGBoost can be found in [41].

SHapley Additive Explanation(SHAP)

Each feature in SHAP is considered a “contributor”. Each sample, when input into the prediction model, will generate a prediction value, and the SHAP value is the marginal contribution of each feature to the prediction result. Assuming that the i^{th} sample is A_i , the j^{th} feature of the i^{th} sample is A_{ij} , the predicted value of the model for this sample is P_i , and the base value of the whole model (usually the mean of the target variables of all samples) is P_{base} , then the formula for calculating the SHAP value is as follows.

$$P_i = P_{base} + f(A_{i1}) + f(A_{i2}) + \dots + f(A_{ij}) \quad (3)$$

where, $f(A_{ij})$ is the SHAP value of A_{ij} . If $f(A_{i1}) > 0$, it indicates that the feature contributes positively to the predicted value and drives the predicted value up. On the contrary, if $f(A_{i1}) < 0$, it indicates that the feature will drive the predicted value to decrease and has a negative effect. The biggest advantage of SHAP value over the traditional feature selection method is that it can reflect the magnitude of each feature’s impact in each sample and also show the positive and negative impact. The detailed process of SHAP can be found in [42].

K Best

The Select K Best method is to select the best K features among all the features to form a new feature set. The algorithm calculates a set of data with scores and p values based on the score function f and then sorts them to select the features with the highest scores or lowest p values. The scoring function f is as follows.

Assuming H_0 : The means of the overall sample of different categories are not significantly different, $\mu_{S_+} = \mu_{S_-}$. H_1 : The means of the overall sample differed significantly across categories, $\mu_{S_+} \neq \mu_{S_-}$.

$$f = \frac{\frac{s_A}{r-1}}{\frac{s_E}{n-r}} \quad (4)$$

where s_A and s_E are the between-group and within-group deviations, respectively, and f obeys the $F(r-1, n-r)$ distribution. where n is the total number of samples and r is the overall number. S_+ is the set of values when the sample belongs to the positive class. S_- is the set of values when the sample belongs to the negative class.

When statistics f is relatively large, that is, over $F_{1-\alpha}(r-1, n-r)$, the more certain we are to reject H_0 , that is, the more helpful the feature is in predicting the category. The detailed process of K Best can be found in [35].

Artificial Neural Network (ANN)

ANN [30] simulates the operation of a nerve cell in a living organism by forming a network structure with hierarchical, connected neurons. The ANN model mainly consists of an input layer, an output layer, and a hidden layer, and there can be more than one hidden layer sandwiched between the input and output layers. The simplest ANN model contains only one hidden layer. Suppose the number of layers is K . The number of neurons in each layer from the input layer to the output layer is $m_0, m_1, \dots, m_k$, and the output vector of each layer of the network is:

$$\text{Input layer : } I^{(0)} = [I_1^{(0)}, I_2^{(0)}, \dots, I_{m_0}^{(0)}]^T \quad (5)$$

$$\text{Hidden layer 1 : } I^{(1)} = [I_1^{(1)}, I_2^{(1)}, \dots, I_{m_1}^{(1)}]^T \quad (6)$$

$$\text{Hidden layer 2 : } I^{(2)} = [I_1^{(2)}, I_2^{(2)}, \dots, I_{m_2}^{(2)}]^T \quad (7)$$

⋮

$$\text{Output layer : } I^{(k)} = [I_1^{(k)}, I_2^{(k)}, \dots, I_{m_k}^{(k)}]^T \quad (8)$$

The weight matrix and bias vector and activation function for each layer are:

$$\text{Hidden layer 1 : } W^{(1)}, b^{(1)}, f^{(1)} \quad (9)$$

$$\text{Hidden layer 2 : } W^{(2)}, b^{(2)}, f^{(2)} \quad (10)$$

⋮

$$\text{Output layer : } W^{(k)}, b^{(k)}, f^{(k)} \quad (11)$$

The function $f^{(i)}$ can be the commonly used sigmoid function or tanh function.

Thus, the derivation of the output expression for the k^{th} layer is:

$$net_i^{(k)} = \sum_{j=1}^{m_{k-1}} W_{ij}^{(k)} I_j^{(k-1)} + b_i^{(k)}, (1 \leq i \leq m_k) \quad (12)$$

$$net^{(k)} = W^{(k)} I^{(k-1)} + b^{(k)} = [net_1^j, net_2^j, \dots, net_{m_k}^j]^T \quad (13)$$

$$I^{(k)} = f^{(k)} net^{(k)} = [I_1^{(j)}, I_2^{(j)}, \dots, I_{m_k}^{(j)}]^T \quad (14)$$

Bayesian Optimization (BO)

The neural network model is considered a black box model, and the specific form of the objective function is not available. Bayesian optimization replaces the objective function by a proxy function. Based on the constructed proxy function, more points can be collected near the possible minimum to update the proxy function, which makes it closer to the shape of the real objective function and easier to find the minimum value of the objective function. Bayesian optimization uses the previously observed prior knowledge of historical information at each iteration to perform the next optimization. Repeating the above process, the minimum point of the function can be found eventually. Algorithm 1 gives the specific procedure of Bayesian optimization. More detailed information can be found in [43].

Algorithm 1: Bayesian optimization

Initialize: the prior distribution of the substitution function
Input: f model, χ hyperparametric search space, S Acquisition Function, M the model obtained by fitting to the data set D
 $D \leftarrow \text{InitSamples}(f, \chi)$
For $i \leftarrow |D|$ **to** T **do**
 $p(y|x, D) \leftarrow \text{fit model}(M, D)$
 $x_i \leftarrow \text{argmax}_{x \in \chi} S(x, p(y|x, D))$
 $y_i \leftarrow f(x_i)$
 $D \leftarrow D \cup (x_i, y_i)$
end for

IMFCE-BO-ANN for Stock Trading Signal Prediction

Interpretable Embedded Multicriteria Feature Cross-selection Engineering (IMFCE)

High-dimensional data often has a large number of redundant features. If all the data is input into the model for training without selection, it will not only increase the computational burden but also decrease the accuracy of the model. Therefore, it is especially important to filter the features. Most of the existing financial time series studies have selected only one individual feature selection method, and the selected ones will be left and other features will be removed. The feature set data is likely to vary with the change of feature selection method, so the selected features cannot adequately represent the data information and reduce the performance of the model. Therefore, a feature

cross-selection engineering based on multiple feature selection criteria, Multicriteria Feature Cross-selection Engineering, is proposed here. The feature engineering incorporates three different types of feature selection algorithms, LightGBM, K Best, and SHAP values based on XGBoost. LightGBM embeds feature selection into the model construction. The attributes of the features evaluated according to the model itself are used as evaluation metrics, and their ranking is performed. The SHAP value based on XGBoost combines the Shapley value of cooperative game theory with XGBoost and uses the absolute value of SHAP of each feature on the overall sample to take the average value to represent the importance of the feature, so the larger the mean SHAP value, the more important the feature is. K Best belongs to the traditional filtered feature selection algorithm, which calculates and ranks the statistical metrics of each feature and selects the top K features. Algorithm 2 shows the detailed process of Multicriteria Feature Cross-selection Engineering.

Algorithm 2: IMFCE model

Input: Training set: D_{train} , Testing set: D_{test}

1. Input the Training set into the K Best model.
2. Ranking features based on scores.
3. Save the first k features to Feature set₁.
4. Input the Training set into the LightGBM model.
5. Train the LightGBM model.
6. Save the top k features with high feature importance values to Feature set₂.
7. Input the Training set into the XGBoost model.
8. Train the XGBoost model.
9. Interpreting XGBoost with SHapley Additive exPlanation and outputting SHAP values.
10. Save the top k features with high SHAP values to Feature set₃.
11. Final feature set=Feature set₁ $\cap$ Feature set₂ $\cap$ Feature set₃

Output: The Final feature set

Framework of Proposed Model

The proposed model consists of four main stages. (1) Data Acquisition. This phase mainly completes data collection, calculation of technical indicators, classification and labeling of key points of trading signals, and data cleaning and standardization (2) Feature engineering. This phase puts the data through Multicriteria Feature Cross-selection Engineering algorithm and divides the data into training and testing sets with sliding windows. (3) Training phase of BO-ANN model. (4) Prediction and performance evaluation phase. The whole process of the proposed model is shown in Fig. 1.

(1) Data Acquisition

In the first stage, we first collect raw stock trading data, including daily opening, closing, high and low prices of the stock data. Then, the technical indicator data is calculated from the raw stock trading data. Third, this paper uses the algorithm of time series extreme value points to obtain the key points of stock trading signals. In the process of obtaining extreme points, it is very important to set an appropriate order (how many points are used on both sides for comparison). If setting a larger order, the time series extreme points algorithm will produce fewer extreme points and the number of key points of the stock trading signals will be reduced. On

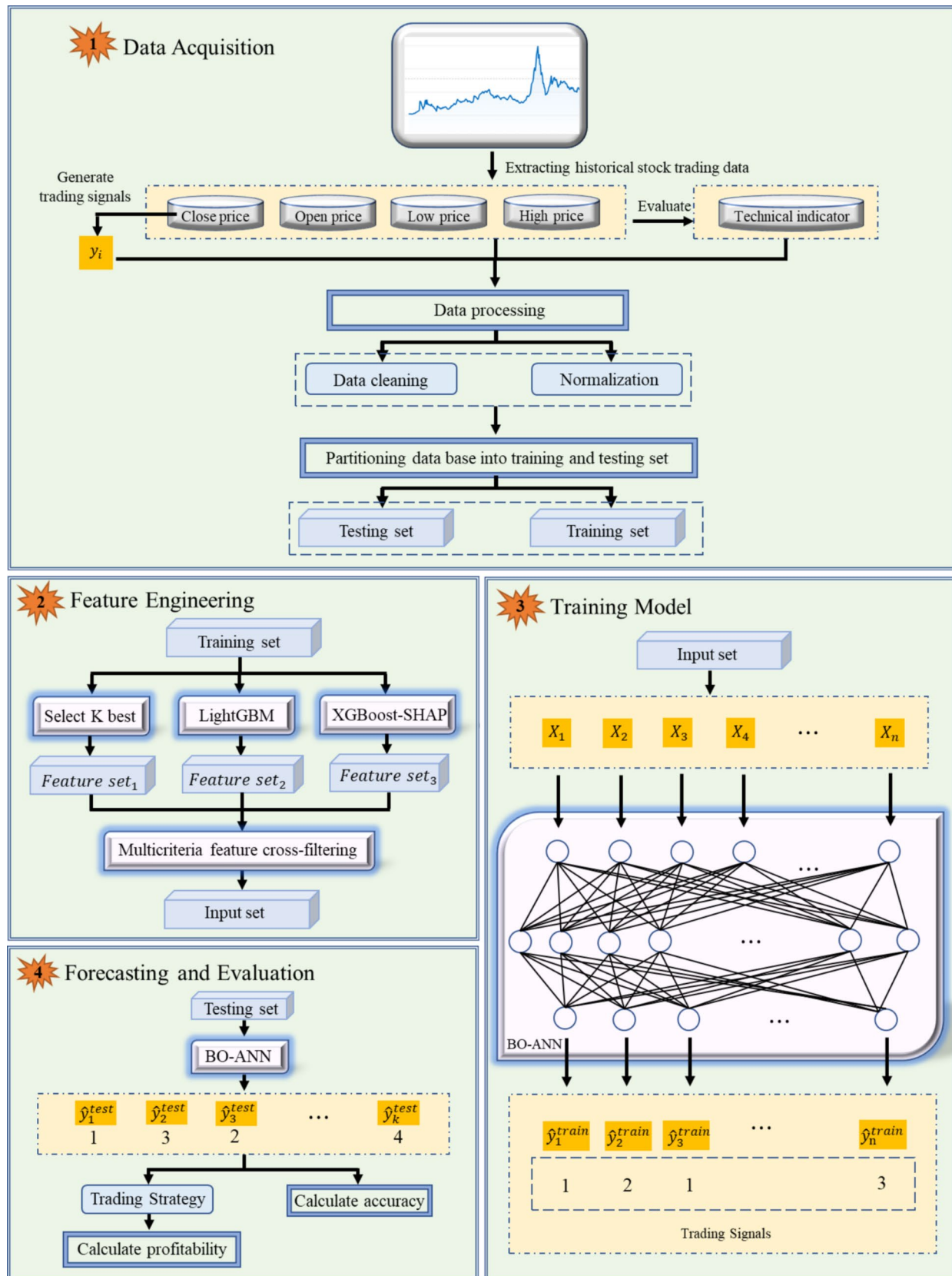


Fig. 1 The framework diagram of proposed model

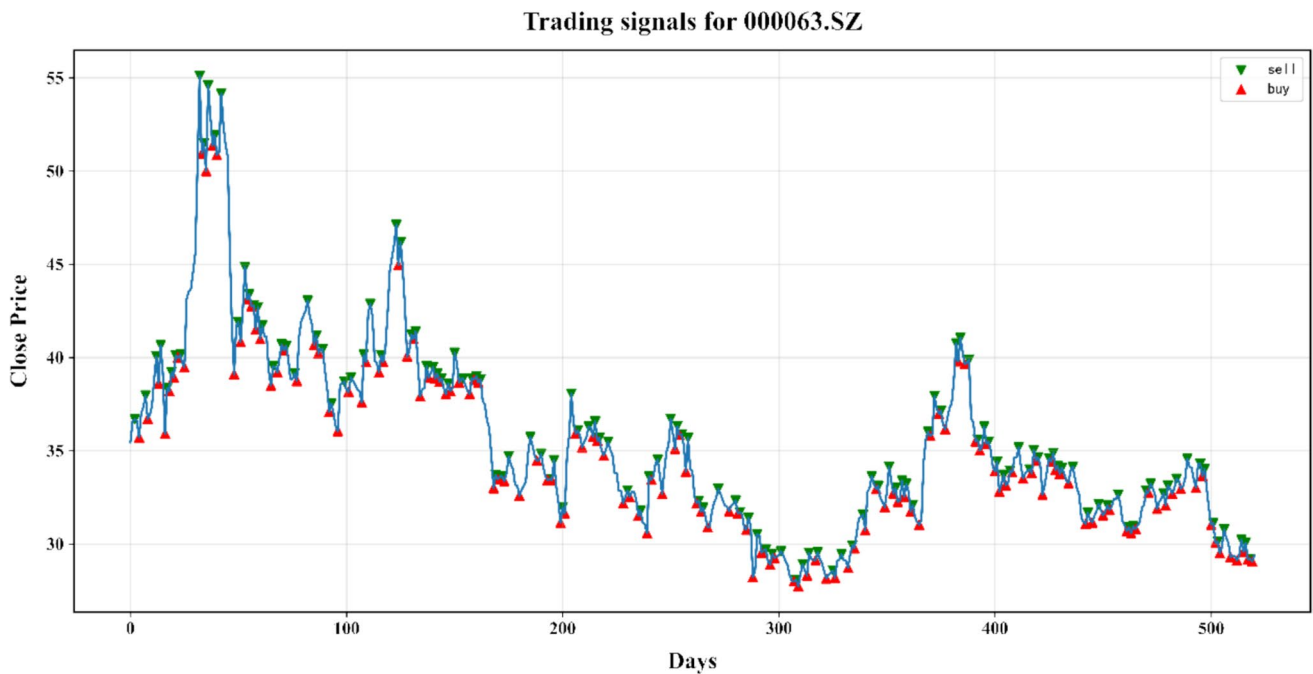


Fig. 2 The trading signal points for a stock (000063.SZ)

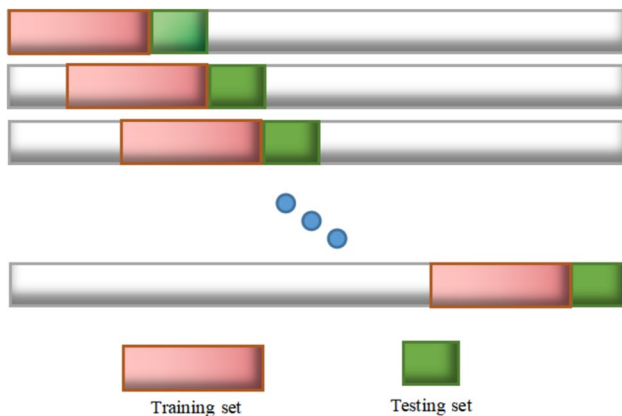


Fig. 3 Training and testing sets with sliding windows

the contrary, if a smaller order is set, more key points of stock trading signals will be generated. Therefore, considering the equilibrium of the quantity of each type in the four types of problems, when order is 1, the quantity of each type of trading point is basically balanced, so order is chosen to be 1. The daily stock prices are processed into four categories according to the classification rules. The extremely large value points are marked as Strongly Recommended Buy Signals (SRBS) and the extremely small value points are marked as Strongly Recommended Sell Signals (SRSS). The remaining points are classified according to the formula

$$\Delta C_i = \frac{C_{i+1} - C_i}{C_i} \times 100, \forall i = 1, 2, 3, \dots, n-1.$$
 If $\Delta C_i > 0$, it is marked as Commonly Recommended Buy Signal (CRBS). If $\Delta C_i < 0$, it is marked as a Commonly Recommended Sell Signal (CRSS). Fig. 2 shows the trading signal points for a stock (000063.SZ), where the horizontal axis represents the trading day in days and the vertical axis represents the daily closing price in RMB. Finally, data cleaning and normalization are performed on the acquired initial dataset.

(2) Feature Engineering

In the second stage, the importance of features in each training set is examined separately using XGBoost-SHAP, LightGBM, and K Best. The feature importance values output by each feature selection method is sorted in descending order, and the top 20 are selected and saved as feature sets, which are recorded as $Featureset_{1 \sim 3}$. Then, the $Featureset_i$ derived from different feature filtering methods are intersected, and the features that all three methods agree have high importance are selected as input features to form the final feature set. Finally, we collect the raw data, the features engineered by feature filtering, and the classification labels in a data matrix, which is then divided into training and test datasets with sliding windows, as shown in Fig. 3.

(3) Training Model

In the third and fourth stages, we treat both the training and test sets of each sliding window as a four-class classification problem, and use ANN classifiers to complete the classification and prediction of key points of stock trading signals. The number of hidden layer neurons, the maximum number of iterations, and the batch size of the ANN model are optimized using Bayesian in the ANN prediction model. After the model hyperparameters are selected, the training set is input into the BO-ANN model for training. Finally, the trained BO-ANN model is used to predict the key points of the stock trading signals in the test set, and the accuracy and returns of the model are calculated.

(4) Forecasting and Evaluation

The trained model is used to predict the key points of the stock trading signals in the test set, and the accuracy and returns of the model are calculated. The accuracy is the number of correctly predicted samples as a percentage of the total, and the specific formula is as follows.

$$Accuracy = \frac{TP + TN}{TP + FN + FP + TN} \quad (15)$$

$$v_{money}(i) = \begin{cases} Need_1^{money} & \text{if } i = 1 \\ v_{money}(i-1) + (Need_i^{money} - b_{money}(i-1)) & \text{if } 0 \leq b_{money}(i-1) \leq Need_i^{money}, i > 1 \\ v_{money}(i-1) & \text{if } b_{money}(i-1) > Need_i^{money}, i > 1 \end{cases} \quad (17)$$

$$b_{money}(i) = \begin{cases} 0 & \text{if } i = 1 \\ b_{money}(i-1) - Need_i^{money} & \text{if } b_{money}(i-1) > Need_i^{money}, i > 1 \\ 0 & \text{if } 0 \leq b_{money}(i-1) \leq Need_i^{money}, i > 1 \end{cases} \quad (18)$$

Selling: If the signal is SRSS and $b_{stock}(i) > 0$, investors will sell all of their shares. When all shares are sold, calculate b_{money} and b_{stock} according to Eqs. (19)–(20).

Strategy 2: Total investment limit: 10,000 CNY. This strategy is suitable for retail investors.

Where True Positive (TP) means positive true value and positive predicted value, False Negative (FN) means positive true value, negative prediction, False Positive (FP) means negative true value and positive predicted value, and True Negative (TN) means negative true value and negative prediction. To test the profitability of the model, the returns of the model will be calculated separately using two investment strategies.

Let b_{stock} denote the number of shares balance, b_{money} denote the balance of funds, v_{money} denote the total investment, Avg denote the average price, i means i^{th} day, c_{buy} denote purchase transaction cost rate, and c_{sell} denote sell transaction cost rate.

Strategy 1: No limit on total investment, continuous buying allowed. This strategy is suitable for institutional investors with more funds.

Buying: If the signal is SRBS, the investors will buy 100 shares with $Need_i^{money} = 100 \times (Avg_i \times (1 + c_{buy}))$ of money. After buying the stock, b_{stock} , v_{money} , and b_{money} are calculated according to Eqs. (16)–(18), respectively.

$$b_{stock}(i) = \begin{cases} 100 & \text{if } i = 1 \\ b_{stock}(i-1) + 100 & \text{if } i > 1 \end{cases} \quad (16)$$

- Buying: If the signal is SRBS, investors buy with the balance. After buying the stock, calculate b_{stock} and b_{money} according to Eqs. (21)–(22).
- Selling: If the signal is SRSS and $b_{stock}(i) > 0$, investors will sell all of their shares. calculate b_{stock} and b_{money} according to Eqs. (19)–(20).

$$b_{money}(i) = b_{money}(i-1) + b_{stock}(i) \times Avg_i \times (1 - c_{sell}) \quad (19)$$

$$b_{stock}(i) = 0 \quad (20)$$

Table 1 Description of data sample

Trend	Stock code	Data range
Uptrend	600018.SH, 600989.SH, 603806.SH, 601985.SH, 000301.SZ	13/Jan/2021 to 25/Feb/2022
Steady trend	600845.SH, 601216.SH, 601155.SH, 002271.SZ	13/Jan/2021 to 25/Feb/2022
Downtrend	600660.SH, 600872.SH, 688036.SH, 002714.SZ, 002568.SZ, 000625.SZ, 300433.SZ, 300015.SZ, 000063.SZ, 300760.SZ, 002032.SZ	13/Jan/2021 to 25/Feb/2022

$$b_{stock}(i) = \begin{cases} 100 \text{if } b_{money}(1) > Need_1^{money}, i = 1 \\ b_{stock}(i-1) + 100 \text{if } b_{money}(i-1) > Need_i^{money}, i > 1 \\ b_{stock}(i-1) \text{if } 0 \leq b_{money}(i-1) \leq Need_i^{money}, i > 1 \end{cases} \quad (21)$$

$$b_{money}(i) = \begin{cases} 10000 - Need_i^{money} \text{if } i = 1 \\ b_{money}(i-1) - Need_i^{money} \text{if } b_{money}(i-1) > Need_i^{money}, i > 1 \\ b_{money}(i-1) \text{if } 0 \leq b_{money}(i-1) \leq Need_i^{money}, i > 1 \end{cases} \quad (22)$$

At the end of the investment period, all shares must be sold at the average price. The profit for strategy is computed by

$$profit_{money} = \frac{b_{money} - v_{money}}{v_{money}} \quad (23)$$

Case Study

Data Acquisition

In this paper, twenty stocks were randomly selected from the Shanghai Stock Exchange and Shenzhen Stock Exchange for the time period January 13, 2021 to February 25, 2022 as experimental data for empirical analysis, with the key point of the trading signal of the closing price of the stock sample as the prediction target. All stock data are obtained through the python financial data interface package Tushare. In the stock market, all stocks can be classified as up trending, steady trending, and down trending stocks according to certain rules. Therefore, this paper classifies the selected 20 stocks based on the closing price change rate between the start and end date of the stock. If the rate of change is over 15%, then the stock is in an uptrend; if the rate of change is less than -15%, then the stock is considered to be in a downtrend; others are stocks with a steady trend. The selected constituent stocks are shown in Table 1. Among the stocks selected for this paper, most of them are in a downtrend, four are in a steady trend, and five are in an uptrend during the same time period. The factors that need to be considered in stock trading signal forecasting are complex. Recently, several studies have shown that technical indicators can provide more information than raw trading data in financial time series forecasting. To obtain more comprehensive data information in this paper, the historical closing price of each stock is used as input characteristics by a series of mathematical formulas to obtain the corresponding technical indicators. These technical indicator data are divided into five main categories of 37 technical indicators, and Table 2 provides more information on the technical indicators.

Normalized Processing

Considering that stock-related data usually have large differences in values, this not only increases the computational effort of the data, but also features with smaller values may be dominated by individual features with larger values. Therefore, normalizing the data is a key step before doing feature selection.

The normalized formula is as follows.

$$\hat{S}_{ij} = \frac{S_{ij} - S_{minj}}{S_{maxj} - S_{minj}} \quad (24)$$

Feature Engineering

In this experiment, the entire stock history trading data 250:30 is divided into overlapping training and testing sets, where the amount of data in the training set is 250 and the amount of data in the testing set is 30. In each training set, the features are filtered using Multicriteria Feature Cross-selection Engineering. The initial feature set is tested for feature importance by three different feature selection models, XGBoost-SHAP, LightGBM, and K-best, respectively, and then the features are ranked in descending order of importance, and the top 20 features constitute a feature subset. The results of feature importance calculated by three different feature selection models XGBoost-SHAP, LightGBM, and K-best are shown in Table 3. Due to space constraints, we only show the S_{train}^1 with an uptrend. Model parameter setting for XGBoost-SHAP and LightGBM: n_estimators is 100. Model parameter setting for XGBoost-SHAP: k='all'. A total of three feature subsets are formed according to this method. Finally, the three different feature subsets are cross-selected, and the same features belonging to $Featureset_1$, $Featureset_2$, and $Featureset_3$ constitute the final input feature set. Tables 4, 5, 6 show the number of features selected by IEMFCE for each stock.

Building Predictive Models

Next, an appropriate trading signal prediction model is built and trained for each training and testing set. The parameters maximum iterations, number of hidden neurons, and batch size of the ANN model used in this paper were chosen using a Bayesian optimization algorithm based on 15-fold cross-validation. In the experiments, all stock trading signal prediction models were implemented on a personal computer with Intel Core i5 CPU, 2.42 GHz using jupyter notebook. The tool used to build the model is Tensorflow 2.7.0. Figs. 4 and 5 show the buy and sell signals generated by IEMFCE-BO-ANN on a test set of stocks with different trends. In the figures, the red upper triangle pattern represents the buy

Table 2 The technical indicators

Category	Technical indicators	Explanation
Overlap Studies	Double Exponential Moving Average (DEAM)	$DEMA = 2 \times EMA - EMA(EMA)$
	Kaufman Adaptive Moving Average (KAMA)	$CurrentKAMA = PriorKAMA + SC \times (Price - PriorKAMA)$ $SC(SmoothingConstant) = [ER \times (fastestSC - slowestSC) + slowest]^2$ $ER = \frac{Change}{Volatility}$ $Change = Close - Close(nperiodsago) $ $Volatility = \frac{sum Close - PriorClose }{n_1 + 1}$ $fastestSC = \frac{2}{n_1 + 1}$ $slowestSC = \frac{2}{n_2 + 1}$
	Moving Average (MA)	$MA(n) = \frac{1}{n} \sum_{i=1}^n C_i$
	Exponential Moving Average (EMA)	$EMA(n) = \begin{cases} C_1 & \text{if } n = 1 \\ \frac{2}{n+1} * C_n + \frac{n-1}{n+1} * EMA(n-1) & \text{if } n > 1 \end{cases}$
	Simple Moving Average (SMA)	$SMA = \frac{\sum_{i=1}^n price}{n}$
	Triple Exponential Moving Average (TEMA)	$TEMA = 3 \times EMA - 3 \times EMA(EMA) + EMA(EMA(EMA))$
	Triangular Moving Average (TRIMA)	If n is even: $TMA = \frac{n}{2} periodSMAof\left(\frac{n}{2} + 1 periodSMAof(price)\right)$ If n is odd: $TMA = \frac{n+1}{2} periodSMAof\left(\frac{n+1}{2} periodSMAof(price)\right)$
Momentum Indicators	Weighted Moving Average (WMA)	$WMA = \frac{price \times n + price_{[-1]} \times (n-1) + price_{[-2]} \times (n-2) \dots price_{[-(n-1)]} \times 1}{n \times (n+1) / 2}$
	Average Directional Movement Index (ADX)	$ADX = \frac{ADX_{[-1]} \times (n-1) + DX}{n}$
	Average Directional Movement Index Rating (ADXR)	$ADXR = \frac{ADX_{[0]} + ADX_{[-n]}}{2}$
	Absolute Price Oscillator (APO)	$PO = SlowMA(price) - FastMA(price)$
	Aroon Oscillator (AROONOSC)	$AroonOscillator = AroonUp - AroonDown$ $AroonUp = 100 \times \left(\frac{n - PeriodsSinceHighsHigh}{n} \right)$ $AroonDown = 100 \times \left(\frac{n - PeriodsSinceLowestLow}{n} \right)$
	Chande Momentum Oscillator (CMO)	If $price > price_{[-1]}$ then: $up = price - price_{[-1]}$ $down = 0$ If $price < price_{[-1]}$ then: $up = 0$ $down = price_{[-1]} - price$ $ups_i = \sum_{i=1}^n up$ $downs_i = \sum_{i=1}^n down$ $CMO = 100 \times \frac{(ups - downs)}{(ups + downs)}$
	Momentum Index (MOM)	$MoM(i, n) = C_i - C_{i-n}$
	Relative Strength Index (RSI)	$RSI(n) = 100 - \frac{100}{1 + \frac{EMA(n)_{up}}{EMA(n)_{down}}}$ $EMA(n)_{up}$ is upward change, $EMA(n)_{down}$ is downward change form.
	Rate of change (ROC)	$((price / prevPrice) - 1) * 100$
	Rate of change Percentage (ROCP)	$(price - prevPrice) / prevPrice$
	Rate of change ratio (ROCR)	$(price / prevPrice)$
	Williams' %R (WILLR)	$\%R = 100 * \frac{HighestHigh(lastnperiods) - close}{HighestHigh(lastnperiods) - LowestLow(lastnperiods)}$
Volume Indicators	On Balance Volume (OBV)	If $close > close_{[-1]}$ then: $OBV = OBV_{[-1]} + volume$ else if $close < close_{[-1]}$ then: $OBV = OBV_{[-1]} - volume$ else : $OBV = OBV_{[-1]}$
	Chaikin A/D Line (AD)	$CLV = \left(\frac{(close - low) - (high - close)}{(high - low)} \right)$ $AD = AD_{[-1]} + CLV \times volume$
Volatility Indicators	Average True Range (ATR)	$ATR = \frac{TR_{[-1]} \times (n-1) + TR}{n}$ $TR = TrueHigh - TrueLow$ $TrueHigh = Highestofhigh_{[0]} or close_{[-1]}$ $TrueLow = Lowestoflow_{[0]} or close_{[-1]}$
	True Range (TRANGE)	$TR = TrueHigh - TrueLow$ $TrueHigh = Highestofhigh_{[0]} or close_{[-1]}$ $TrueLow = Lowestoflow_{[0]} or close_{[-1]}$
Price Transform	Weighted Close Price (WCLPRICE)	$WeightedClose = \frac{high + low + close \times 2}{4}$

Table 3 Results of the top 20 features' importance calculated for S^1_{train} with an uptrend

Stock code	Results of the top 20 features	
600018.SH	XGBoost-SHAP	Price_change, WILLR, RSI10, OBV, ATR, APO, volume, TRANGE, AD, DEMA30, RSI30, TEMA10, TEMA30, ROC30, CMO, MOM30, ROC10, AROONOSC, MOM10, TRIMA10
	LightGBM	Price_change, OBV, WILLR, RSI10, APO, AD, ATR, volume, ADXR, AROONOSC, TRANGE, MOM30, MOM10, CMO, RSI30, ADX, ROC30, KAMA30, WCLPRICE, ROC10
	K-best	WILLR, Price_change, RSI10, CMO, MOM10, ROC10, ROCP10, volume, RSI30, AROONOSC, TRANGE, MOM30, ROCP30, ROC30, KAMA30, WCLPRICE, OBV, AD, TRIMA30, SMA30
600989.SH	XGBoost-SHAP	Price_change, WILLR, RSI10, APO, TRANGE, AD, MOM30, MA10, ADX, CMO, ROC10, volume, MOM10, KAMA30, RSI30, TEMA30, ADXR, TRIMA30, MA30, TRIMA10
	LightGBM	Price_change, WILLR, RSI10, AD, APO, MOM10, TRANGE, CMO, volume, MOM30, ADX, ADXR, ROC10, RSI30, DEMA10, ROC30, ATR, AROONOSC, WCLPRICE TEMA10
	K-best	WILLR, RSI10, Price_change, CMO, ROC10, ROCP10, MOM10, RSI30, volume, AROONOSC, WCLPRICE, TRANGE, TEMA10, ROC30, ROCP30, MOM30, DEMA10, OBV, AD, WMA10
603806.SH	XGBoost-SHAP	Price_change, WILLR, APO, RSI10, TRANGE, c, CMO, ROC10, ADXR, OBV, MOM10, MA30, AROONOSC, WCLPRICE, ROC30, RSI30, ADX, TRIMA30, AD, TRIMA10
	LightGBM	Price_change, WILLR, volume, MOM10, ROC10, TRANGE, APO, OBV, ADXR, CMO, AROONOSC, RSI10, ADX, ATR, RSI30, AD, ROC30, MOM30, MA30, TRIMA30
	K-best	WILLR, Price_change, RSI10, ROC10, ROCP10, CMO, MOM10, AROONOSC, RSI30, MOM30, ROC30, ROCP30, TRIMA30, APO, SMA30, MA30, WMA30, EMA30, KAMA30, DEMA30
601985.SH	XGBoost-SHAP	Price_change, WILLR, RSI10, APO, AROONOSC, ADX, CMO, TRANGE, volume, ATR, ROC30, OBV, AD, RSI30, ROC10, TRIMA30, TEMA10, TRIMA10, MA10, ADXR
	LightGBM	Price_change, WILLR, RSI10, APO, ADX, ROC30, AROONOSC, RSI30, CMO, ADXR, volume, ROC10, MOM30, MOM10, ATR, AD, TRANGE, WCLPRICE, TRIMA30, TEMA10
	K-best	WILLR, RSI10, Price_change, CMO, MOM10, ROCP10, ROC10, RSI30, volume, AROONOSC, AD, MOM30, ROCP30, ROC30, WCLPRICE, TEMA10, OBV, TRANGE, DEMA10, APO
000301.SZ	XGBoost-SHAP	Price_change, WILLR, RSI10, ROC10, volume, APO, ADX, AROONOSC, MOM10, CMO, ADXR, TRANGE, ATR, MOM30, OBV, WCLPRICE, AD, TRIMA30, DEMA30, RSI30
	LightGBM	Price_change, WILLR, APO, RSI10, volume, ROC10, ADX, MOM10, ATR, TRANGE, ADXR, OBV, AD, MOM30, CMO, WCLPRICE, RSI30, ROC30, AROONOSC, MA10
	K-best	WILLR, RSI10, CMO, Price_change, ROC10, ROCP10, MOM10, AROONOSC, RSI30, volume, TRANGE, ADXR, TRIMA30, SMA30, MA30, ROC30, ROCP30, WMA30, MOM30, EMA30

Table 4 Feature selection results for stocks with an uptrend

Stock code	The number of features for each training set with sliding windows								
	S^1_{train}	S^2_{train}	S^3_{train}	S^4_{train}	S^5_{train}	S^6_{train}	S^7_{train}	S^8_{train}	S^9_{train}
600018.SH	12	14	12	12	12	13	14	12	13
600989.SH	11	15	14	15	14	15	12	12	10
603806.SH	12	10	8	9	12	15	14	12	15
601985.SH	13	15	12	16	15	15	10	11	14
000301.SZ	14	17	11	13	15	15	14	14	

Table 5 Feature selection results for stocks with a steady trend

Stock code	The number of features for each training set with sliding windows								
	S^1_{train}	S^2_{train}	S^3_{train}	S^4_{train}	S^5_{train}	S^6_{train}	S^7_{train}	S^8_{train}	S^9_{train}
600845.SH	12	13	12	12	11	10	9	9	11
601216.SH	10	10	10	9	10	14	14	12	10
601155.SH	11	13	12	14	9	11	11	9	12
002271.SZ	10	9	14	14	11	11	9	10	9

Table 6 Feature selection results for stocks with a downtrend

Stock code	The number of features for each training set with sliding windows								
	S^1_{train}	S^2_{train}	S^3_{train}	S^4_{train}	S^5_{train}	S^6_{train}	S^7_{train}	S^8_{train}	S^9_{train}
600660.SH	17	17	12	12	12	12	14	13	13
600872.SH	11	11	11	12	15	13	16	15	13
688036.SH	13	15	12	16	15	15	10	11	14
002714.SZ	13	15	12	11	13	15	17	13	13
002568.SZ	12	16	14	16	14	16	15	13	12
000625.SZ	17	12	11	11	12	11	10	11	14
300433.SZ	14	14	10	8	11	11	15	14	16
300015.SZ	16	12	13	15	13	12	17	14	-
000063.SZ	12	14	15	13	11	12	9	10	11
300760.SZ	13	13	14	14	11	12	12	11	14
002032.SZ	12	10	10	11	14	14	15	14	14

signal, and the green lower triangle pattern represents the sell signal.

Comparison Experiments

For a more comprehensive and quantitative understanding of the performance of the proposed models, five comparative models were selected for comparison, which are ANN, SVM, CNN models alone and SVM model after Bayesian optimization (BO-SVM) and CNN model after putting Bayesian optimization (BO-CNN). Individual ANN, SVM, CNN model parameters are determined using the traditional grid search method. In the BO-CNN model, the traversal method is used to determine the number of convolutional layers, the number of pooling layers, and the number of fully connected layers, and then the Bayesian optimization method based on 15-fold cross-validation is used to determine the number of neurons in each layer. The activation function used for the convolutional and fully connected layers is Relu, while the activation function for the output layer is SoftMax. In the BO-SVM model, Bayesian optimization based on 15-fold cross-validation is used to select the hyperparameters of the SVM.

In this section, the prediction performance of the IEMFCE-BO-ANN algorithm for the key points of trading signals is tested and analyzed. Two trading strategies observe the gains obtained by the model, and then the experimental results are compared with those of IEMFCE-ANN, IEMFCE-BO-SVM, IEMFCE-SVM, IEMFCE-BO-CNN, IEMFCE-CNN, and BHS. Table 7 shows the stock trading parameters in the proposed model. The size of the training set N_{train} is 250 and the test set N_{test} is 30, and the transaction cost is 0.00302 when buying a stock and 0.00402 when selling a stock. Tables 8–10 show the comparison results for stocks in uptrend, steady trend, and downtrend. N_{buy} is the number of selling actions. N_{sell} is the number of buying actions.

As shown in Table 8, comparative experiments were conducted using different forecasting models in each of the five stocks with an upward trend, and the results showed that the proposed model had the best performance. Compared with IEMFCE-ANN, IEMFCE-BO-SVM, IEMFCE-SVM, IEMFCE-BO-CNN, IEMFCE-CNN, and BHS, the model proposed, IEMFCE-BO-ANN, has the highest accuracy, and the accurate rate was 54.83%. The average accuracy of the IEMFCE-ANN, IEMFCE-BO-SVM, IEMFCE-SVM, IEMFCE-BO-CNN, and IEMFCE-CNN models is 51.83%, 47.71%, 45.95%, 46.17 %, and 42.86%, respectively. IEMFCE-BO-ANN model is 7.8% more accurate than other models on average. For stock trading strategy 1, the highest average return obtained is the IEMFCE-SVM model, with an average return of 23.41%. Followed by the IEMFCE-BO-ANN model, the investment income was 15.27%. The reason why IEMFCE-BO-ANN in investment strategy 1 has lower investment returns than the IEMFCE-SVM model, but the classification accuracy is higher than the IEMFCE-SVM model is the model classification effect is not ideal, no matter the stock price is low or at the high level, the number of trading signals predicted by the model is small, and there is no frequent buying and selling. At the same time, the model changes greatly in the observation interval, which causes the phenomenon that although the model accuracy rate is low, the income is high. For strategy 2, the return of the IEMFCE-BO-ANN model is slightly higher than that of the IEMFCE-BO-SVM. We can conclude that the proposed model is effective for stocks with stable trends and can achieve better investment returns. We can get a conclusion that the proposed model is effective in stocks with uptrend and can get better returns.

In Table 9, the performance of IEMFCE-BO-ANN is presented as the best among the comparative experiments with stocks with a steady stock price trend. Compared with IEMFCE-ANN, IEMFCE-BO-SVM, IEMFCE-SVM, IEMFCE-BO-CNN, IEMFCE-CNN, and BHS, IEMFCE-BO-ANN

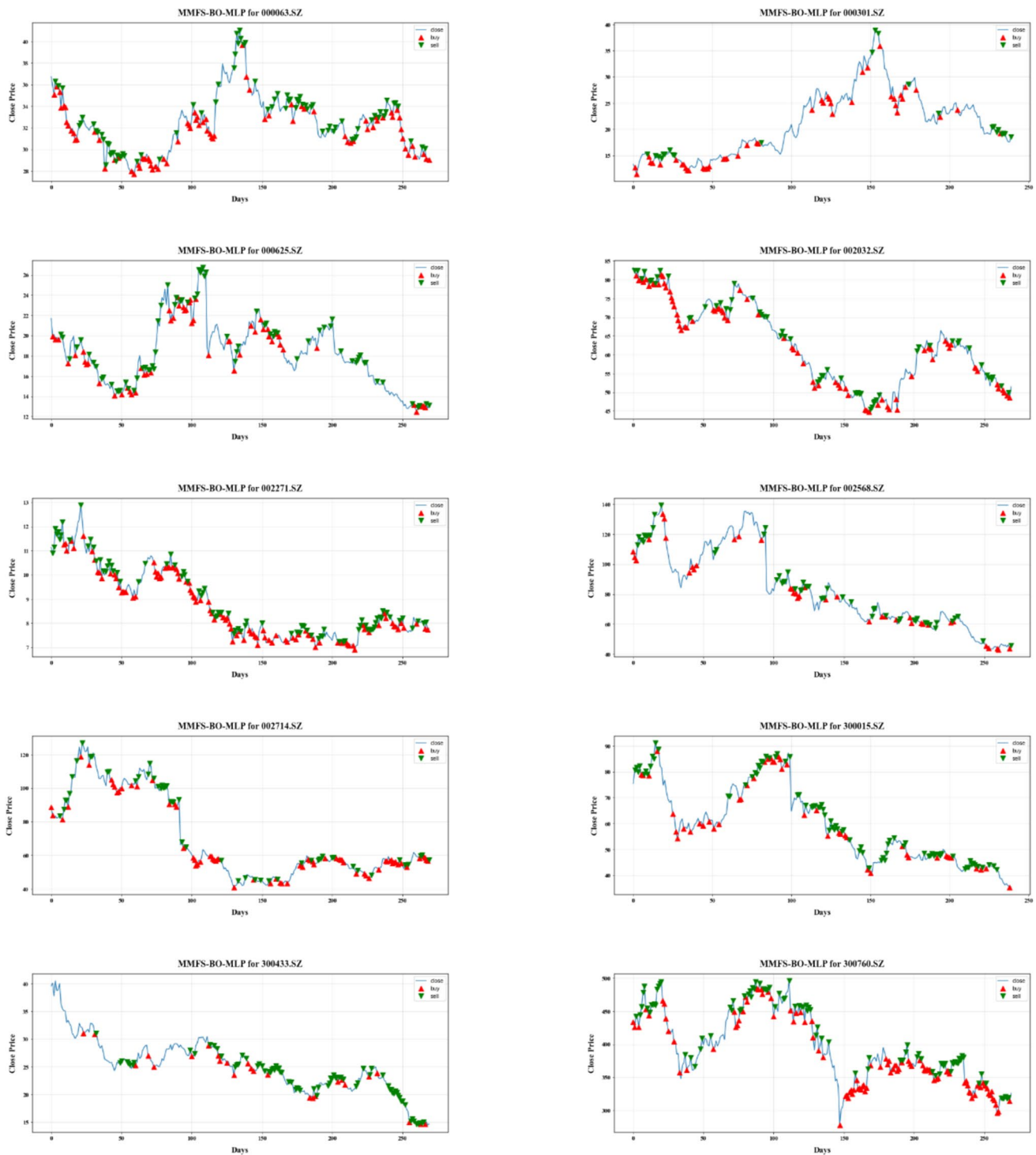


Fig. 4 Buy and sell signals generated by IEMFCE-BO-ANN for the Shenzhen Stock Exchange

has the highest accuracy and the accurate rate is 55.46%. The average accuracy of the IEMFCE-ANN, IEMFCE-BO-SVM, IEMFCE-SVM, IEMFCE-BO-CNN, and IEMFCE-CNN

model is 51.57%, 48.15%, 47.50%, 46.58%, and 43.05%, respectively. The average accuracy of IEMFCE-BO-ANN model is 3.89% higher than IEMFCE-ANN, 7.96% higher

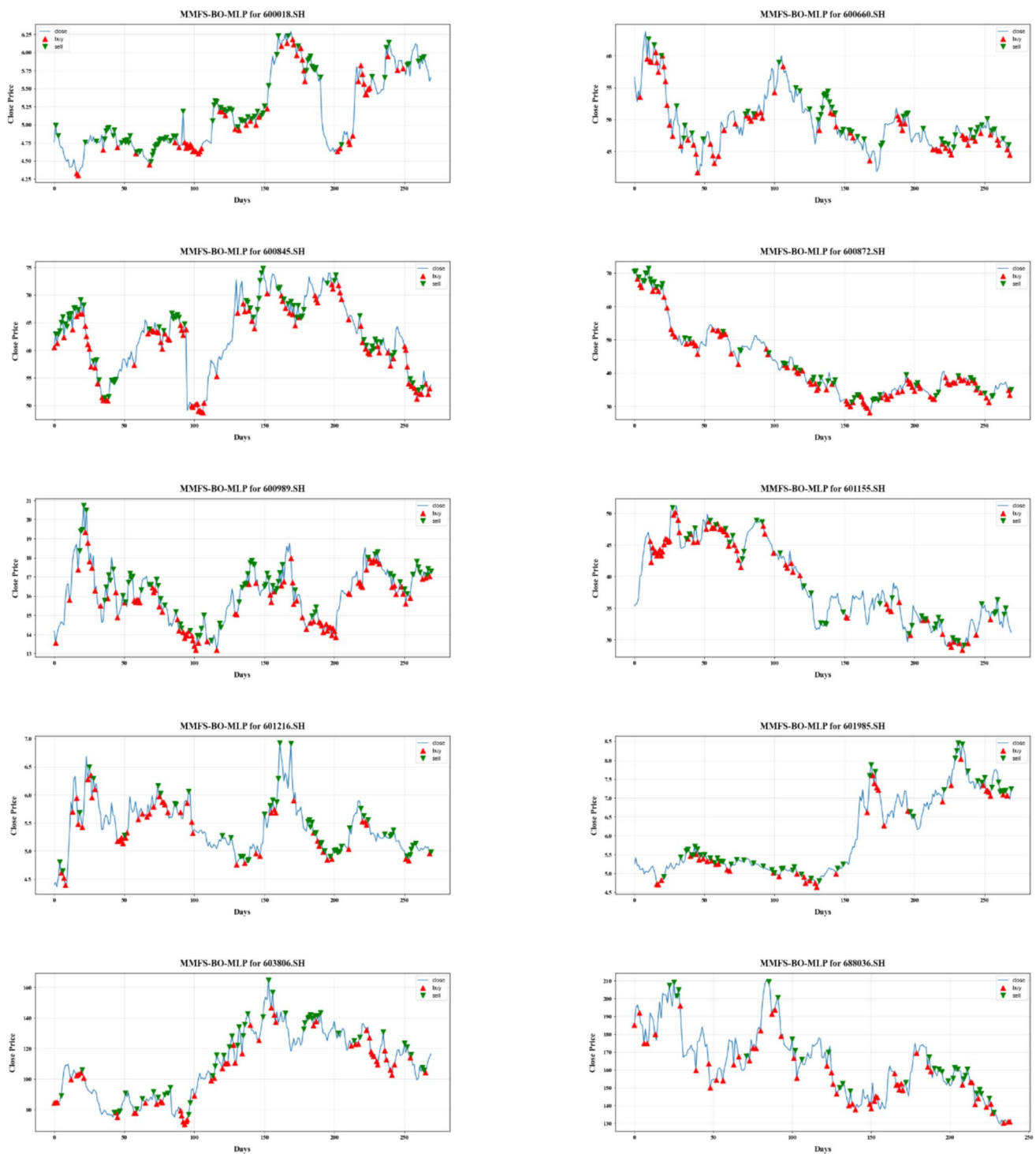


Fig. 5 Buy and sell signals generated by IEMFCE-BO-ANN for the Shanghai Stock Exchange

Table 7 Description of parameters used for stock trading signal prediction

N_{train}	N_{test}	c_{buy}	c_{sell}
250	30	0.00302	0.00402

than IEMFCE-SVM model, 7.31% higher than IEMFCE-BO-SVM, 8.88% higher than IEMFCE-BO-CNN, and higher than IEMFCE-CNN is 12.41% higher and on average 8.09%

Table 8 The comparison results for stocks with an uptrend

Stock code	Method	Accuracy	Strategy1				Strategy2				BHS	
			$v_{money}(CNY)$	Profit	N_{buy}	N_{sell}	$v_{money}(CNY)$	Profit	N_{buy}	N_{sell}	Profit	
600018.SH	MMSF-ANN	50.74%	5180.9	-2.52%	47	20	10,000	-1.31%	47	20	9.18%	
	MMSF-BO-ANN	54.07%	5791.49	5.55%	55	25	10,000	3.21%	55	25	9.07%	
	MMSF-SVM	42.59%	6083.27	4.48%	46	14	10,000	2.72%	46	14	7.11%	
	MMSF-BO-SVM	45.19%	5246.61	-5.33%	33	12	10,000	-2.8%	33	12	0.77%	
	MMSF-CNN	41.48%	5043.28	-18.68%	42	17	10,000	-9.42%	42	17	7.79%	
	MMSF-BO-CNN	45.56%	5112.57	-5.52%	62	24	10,000	-2.82%	62	24	6.97%	
600989.SH	MMSF-ANN	55.56%	41,250.71	14.37%	88	35	10,000	16.48%	68	35	8.52%	
	MMSF-BO-ANN	58.89%	30,236.15	10.35%	89	36	10,000	4.93%	76	36	8.88%	
	MMSF-SVM	52.59%	28,175.32	13.97%	98	32	10,000	7.47%	82	32	9.42%	
	MMSF-BO-SVM	52.96%	26,972.12	15.16%	72	23	10,000	4.56%	60	23	7.91%	
	MMSF-CNN	46.30%	18,499.49	-6.45%	87	35	10,000	-16.40%	78	35	8.83%	
	MMSF-BO-CNN	50.00%	23,527.29	-1.18%	107	38	10,000	-17.19%	82	38	8.53%	
603806.SH	MMSF-ANN	51.85%	71,978.85	-6.99%	71	30	10,000	-17.89%	11	11	12.27%	
	MMSF-BO-ANN	53.70%	82,378.65	16.12%	56	25	10,000	37.86%	15	14	9.82%	
	MMSF-SVM	47.78%	109,002.81	-14.68%	52	16	10,000	-15.69%	6	6	9.73%	
	MMSF-BO-SVM	48.15%	65,969.09	-27.07%	46	16	10,000	-9.25%	5	5	13.33%	
	MMSF-CNN	41.48%	94,519.17	-14.59%	75	24	10,000	-22.70%	4	4	18.57%	
	MMSF-BO-CNN	44.44%	90,060.05	-14.25%	77	23	10,000	-24.05%	4	4	18.55%	
601985.SH	MMSF-ANN	51.85%	3679.77	1.19%	43	24	10,000	0.44%	43	24	16.12%	
	MMSF-BO-ANN	56.67%	3005.71	-4.60%	39	26	10,000	-1.38%	39	26	17.39%	
	MMSF-SVM	42.22%	4779.53	-4.58%	37	18	10,000	-2.19%	37	18	18.06%	
	MMSF-BO-SVM	45.56%	4554.52	-1.68%	48	23	10,000	-0.77%	48	23	19.59%	
	MMSF-CNN	42.96%	4687.72	-12.87%	56	30	10,000	-6.03%	56	30	17.86%	
	MMSF-BO-CNN	46.67%	2704.50	-21.91%	57	36	10,000	-5.93%	57	36	24.23%	
000301.SZ	MMSF-ANN	49.17%	25,145.64	16.35%	44	11	10,000	13.63%	33	11	-11.79%	
	MMSF-BO-ANN	50.83%	24,908.70	48.91%	46	9	10,000	64.72%	29	9	-10.62%	
	MMSF-SVM	44.58%	25,591.21	116.84%	21	4	10,000	114.37%	12	4	3.27%	
	MMSF-BO-SVM	46.67%	28,659.42	44.89%	30	8	10,000	75.02%	20	8	-6.39%	
	MMSF-CNN	42.08%	25,037.14	44.88%	56	6	10,000	59.96%	33	6	-23.59%	
	MMSF-BO-CNN	44.17%	22,768.55	3.01%	68	15	10,000	-4.07%	45	15	-15.83%	
Average	MMSF-ANN	51.83%	29,447.17	4.48%	52	24	10,000	2.27%	40.4	20.2	6.86%	
	MMSF-BO-ANN	54.83%	29,264.14	15.27%	57	24.2	10,000	21.87%	42.8	22	6.91%	
	MMSF-SVM	45.95%	34,726.43	23.21%	50.8	16.8	10,000	21.34%	36.6	14.8	9.52%	
	MMSF-BO-SVM	47.71%	26,280.35	5.19%	45.8	16.4	10,000	13.35%	33.2	14.2	7.04%	
	MMSF-CNN	42.86%	29,557.36	-1.54%	63.2	22.4	10,000	1.08%	42.6	18.4	5.89%	
	MMSF-BO-CNN	46.17%	28,834.59	-7.97%	74.2	27.2	10,000	-10.81%	50	23.4	8.49%	

more accurate than other models. For stock trading strategy 1, the highest average return obtained is the IEMFCE-BO-ANN model, with an average return of 7.76%. For stock trading strategy 2, the highest average return obtained is the IEMFCE-BO-ANN model. So, we can draw a conclusion that the model proposed in this paper is effective in steady trend stocks and can get better returns.

A comparison of 11 declining stocks is presented in Table 10, and the results show that the proposed model has the best performance. Compared with IEMFCE-ANN, IEMFCE-BO-SVM, IEMFCE-SVM, IEMFCE-BO-CNN,

IEMFCE-CNN, and BHS, IEMFCE-BO-ANN has the highest accuracy and the accurate rate is 54.70%. The average accuracy of the IEMFCE-ANN IEMFCE-BO-SVM, IEMFCE-SVM, IEMFCE-BO-CNN, and IEMFCE-CNN model is 51.22%, 47.58%, 47.58%, 44.88%, 46.60%, and 42.76%, respectively. The average accuracy of IEMFCE-BO-ANN model is 3.48% higher than IEMFCE-ANN, 9.82% higher than IEMFCE-SVM model, 7.12% higher than IEMFCE-BO-SVM, 8.1% higher than IEMFCE-BO-CNN, and higher than IEMFCE-CNN is 11.94% higher and on average 8.09% more accurate than other models.

Table 9 The comparison results for stocks with a steady trend

Stock code	Method	Accuracy	Strategy1				Strategy2				BHS	
			$v_{money}(CNY)$	Profit	N_{buy}	N_{sell}	$v_{money}(CNY)$	Profit	N_{buy}	N_{sell}	Profit	
600845.SH	MMSF-ANN	54.81%	54,908.20	-20.99%	98	38	10,000	-45.85%	39	38	-13.52%	
	MMSF-BO-ANN	58.15%	79,662.73	12.50%	90	41	10,000	-38.97%	42	41	-13.35%	
	MMSF-SVM	50.74%	91,867.50	18.05%	104	31	10,000	-40.11%	32	31	-13.07%	
	MMSF-BO-SVM	48.52%	69,221.83	-10.98%	101	31	10,000	-37.68%	32	31	-13.21%	
	MMSF-CNN	42.96%	75,246.77	-24.16%	115	30	10,000	-47.99%	27	27	-13.76%	
	MMSF-BO-CNN	48.15%	67,359.50	-17.57%	112	36	10,000	-45.35%	36	37	-13.79%	
601216.SH	MMSF-ANN	51.48%	6843.56	-0.74%	67	29	10,000	-0.51%	67	29	-9.77%	
	MMSF-BO-ANN	56.67%	3888.64	6.56%	53	27	10,000	2.55%	53	27	-8.70%	
	MMSF-SVM	45.56%	4107.92	-2.52%	41	26	10,000	-1.04%	41	26	-8.99%	
	MMSF-BO-SVM	46.30%	5969.71	-3.93%	35	19	10,000	2.31%	35	19	-11.17%	
	MMSF-CNN	42.96%	10,270.10	-5.05%	113	29	10,000	-5.15%	112	29	-8.34%	
	MMSF-BO-CNN	44.44%	6583.70	-9.22%	105	29	10,000	-6.07%	105	29	-8.92%	
601155.SH	MMSF-ANN	51.11%	27,430.80	-9.48%	53	24	10,000	-27.21%	38	24	-23.82%	
	MMSF-BO-ANN	54.81%	67,402.44	4.59%	65	22	10,000	-19.98%	36	22	-25.27%	
	MMSF-SVM	46.30%	44,438.18	-8.03%	41	12	10,000	-6.72%	24	12	-21.04%	
	MMSF-BO-SVM	47.04%	26,521.70	-8.12%	31	12	10,000	-10.92%	19	12	-23.85%	
	MMSF-CNN	44.81%	50,880.40	-20.78%	62	20	10,000	-55.14%	29	20	-20.23%	
	MMSF-BO-CNN	46.67%	38,980.15	-23.44%	74	25	10,000	-50.87%	35	25	-18.93%	
002271.SZ	MMSF-ANN	48.89%	111,025.28	7.51%	80	32	10,000	-14.10%	42	32	-7.57%	
	MMSF-BO-ANN	52.22%	67,661.75	7.38%	63	21	10,000	2.57%	33	21	-7.16%	
	MMSF-SVM	49.63%	89,460.12	6.66%	78	24	10,000	-6.44%	32	24	-7.70%	
	MMSF-BO-SVM	48.52%	88,129.17	20.35%	60	16	10,000	14.65%	26	16	-7.32%	
	MMSF-CNN	41.48%	79,990.60	-9.64%	78	27	10,000	-46.06%	29	27	-9.75%	
	MMSF-BO-CNN	47.04%	90,807.15	-5.34%	81	29	10,000	-45.76%	31	29	-7.88%	
Average	MMSF-ANN	51.57%	50,051.96	-5.93%	74.5	30.7	10,000	-21.92%	46.5	30.7	-13.67%	
	MMSF-BO-ANN	55.46%	54,653.89	7.76%	67.7	27.7	10,000	-13.46%	41	27.7	-13.62%	
	MMSF-SVM	47.50%	57,468.43	3.54%	66	23.2	10,000	-13.58%	32.2	23.2	12.70%	
	MMSF-BO-SVM	48.15%	47,460.60	-0.67%	56.7	19.5	10,000	-7.91%	28	19.5	-13.89%	
	MMSF-CNN	43.05%	54,096.97	-14.91%	92	26.5	10,000	-38.59%	49.2	25.7	-13.02%	
	MMSF-BO-CNN	46.58%	50,932.63	-13.89%	93	29.8	10,000	-37.01%	51.7	30	-12.38%	

For stock trading strategy 1, the highest average return obtained is the IEMFCE-BO-ANN model, with an average return of -3.52%. Followed by the IEMFCE-BO-SVM model, the return on investment is -9.17%. The average returns of IEMFCE-ANN, IEMFCE-SVM, IEMFCE-BO-CNN, IEMFCE-CNN models are: -10.85%, -12.08%, -16.41%, and -18.54%, respectively. Since the stock is in a downtrend, the yield is negative. For stock trading according to investment strategy 2, the highest average return obtained is the IEMFCE-BO-ANN model, with an investment return of -16.3%. The average investment income of IEMFCE-ANN is the highest, which is: -100.19%, the average investment income of IEMFCE-BO-SVM is: -28.51%, the average investment income of IEMFCE-SVM is: -33.59%, IEMFCE-BO-CNN The average investment return of IEMFCE-CNN is: -44.83%, and the average investment return of IEMFCE-CNN is:

-40.58%. We can conclude that IEMFCE-BO-ANN is effective in stocks in a downtrend and minimizes losses.

Conclusion

In the stock market, where stock transactions are frequent and complex, predicting the movement of stock prices is a very difficult task. This paper proposes an integrated and effective modeling framework that integrates three effective feature engineering methods into a Bayesian-optimized ANN model to predict stock trading signals. The accuracy of the proposed model for predicting stock trading signals is more effective and operable compared to other existing modeling approaches. The model consists of four main phases, data acquisition, feature engineering, model training, and performance evaluation. In the data collection phase, on the

Table 10 The comparison results for stocks with a downtrend

Stock code	Method	Accuracy	Strategy1				Strategy2				BHS	
			$v_{money}(CNY)$	Profit	N_{buy}	N_{sell}	$v_{money}(CNY)$	Profit	N_{buy}	N_{sell}	Profit	
600660.SH	MMSF-ANN	57.04%	30,992.86	-9.14%	63	31	10,000	-22.41%	33	31	-11.60%	
	MMSF-BO-ANN	59.26%	32,537.69	0.48%	66	32	10,000	-20.62%	34	32	-10.27%	
	MMSF-SVM	53.70%	41,853.42	-6.56%	62	23	10,000	-21.64%	24	23	-9.80%	
	MMSF-BO-SVM	55.56%	36,722.04	-9.51%	59	22	10,000	-24.93%	23	22	-10.46%	
	MMSF-CNN	44.44%	47,924.89	-3.16%	57	19	10,000	-13.46%	29	19	-7.18%	
	MMSF-BO-CNN	50.74%	50,858.86	6.62%	78	24	10,000	-17.79%	30	24	-7.05%	
600872.SH	MMSF-ANN	51.48%	69,155.57	-8.91%	89	27	10,000	-64.10%	27	27	-18.48%	
	MMSF-BO-ANN	55.56%	54,564.85	-10.24%	87	30	10,000	-54.53%	30	30	-17.23%	
	MMSF-SVM	47.04%	81,288.38	-2.89%	66	19	10,000	-41.38%	20	19	-20.77%	
	MMSF-BO-SVM	46.67%	99,609.75	-8.07%	76	23	10,000	-54.53%	23	23	-19.61%	
	MMSF-CNN	42.59%	67,096.14	-24.61%	79	30	10,000	-56.19%	31	30	-21.76%	
	MMSF-BO-CNN	43.70%	86,208.32	-19.38%	110	29	10,000	-59.05%	29	29	-19.48%	
688036.SH	MMSF-ANN	51.67%	257,339.65	6.29%	61	17	10,000	-7.415	48	17	-20.54%	
	MMSF-BO-ANN	55.42%	178,153.44	5.00%	54	18	10,000	12.27%	47	18	-19.20%	
	MMSF-SVM	45.00%	147,867.28	-12.80%	46	16	10,000	-16.47%	40	16	-18.94%	
	MMSF-BO-SVM	47.08%	124,410.31	-8.46%	44	16	10,000	-12.09%	42	16	-18.12%	
	MMSF-CNN	48.75%	132,615.29	-26.62%	55	20	10,000	-33.85%	49	20	-23.01%	
	MMSF-BO-CNN	51.25%	173,130.41	-11.46%	79	32	10,000	-28.44%	71	32	-20.22%	
002714.SZ	MMSF-ANN	52.22%	114,061.87	-5.05%	88	28	10,000	-33.41%	23	23	-18.45%	
	MMSF-BO-ANN	56.30%	71,652.92	-1.19%	70	27	10,000	2.76%	29	24	-14.27%	
	MMSF-SVM	42.22%	78,340.84	-11.29%	90	21	10,000	-27.30%	24	21	-15.88%	
	MMSF-BO-SVM	45.56%	97,556.36	0.76%	80	17	10,000	-3.23%	22	17	-15.91%	
	MMSF-CNN	42.96%	133,553.79	-27.57%	86	15	10,000	-42.74%	12	11	-18.49%	
	MMSF-BO-CNN	44.44%	145,096.56	-31.52%	96	27	10,000	-47.88%	13	13	-18.64%	
002568.SZ	MMSF-ANN	47.04%	45,125.96	-33.90%	49	20	10,000	-44.35%	17	17	-44.01%	
	MMSF-BO-ANN	52.59%	77,422.90	-5.21%	39	18	10,000	-0.07%	16	15	-45.62%	
	MMSF-SVM	39.63%	180,957.35	-19.38%	51	9	10,000	-47.18%	10	9	-38.32%	
	MMSF-BO-SVM	44.44%	83,348.04	-16.65%	39	11	10,000	-44.76%	12	11	-38.67%	
	MMSF-CNN	38.52%	98,367.90	-45.84%	66	19	10,000	-55.07%	18	18	-46.92%	
	MMSF-BO-CNN	40.37%	103,252.85	-35.65%	75	25	10,000	-55.42%	16	16	-45.36%	
000625.SZ	MMSF-ANN	52.96%	10,675.67	-20.61%	61	36	10,000	-22.25%	60	36	-27.53%	
	MMSF-BO-ANN	53.70%	9411.44	-13.26%	56	30	10,000	-12.48%	56	30	-30.00%	
	MMSF-SVM	48.52%	19,586.89	-27.82%	66	27	10,000	-52.47%	57	27	-28.12%	
	MMSF-BO-SVM	49.63%	7858.23	-10.35%	50	27	10,000	-8.14%	50	27	-24.83%	
	MMSF-CNN	44.44%	11,624.09	-31.53%	55	25	10,000	-34.19%	53	25	-26.47%	
	MMSF-BO-CNN	47.04%	29,663.03	-18.69%	76	30	10,000	-37.73%	55	30	-26.38%	
300433.SZ	MMSF-ANN	51.85%	14,473.98	-16.07%	41	26	10,000	-21.52%	37	26	-43.58%	
	MMSF-BO-ANN	52.22%	7942.83	-10.18%	26	17	10,000	-8.08%	26	17	-38.20%	
	MMSF-SVM	44.81%	19,601.34	-11.48%	28	13	10,000	-25.16%	24	13	-43.71%	
	MMSF-BO-SVM	47.07%	9444.57	-26.01%	20	12	10,000	-24.57%	20	12	-43.13%	
	MMSF-CNN	45.93%	32,233.64	-14.34%	52	21	10,000	-35.34%	30	21	-43.41%	
	MMSF-BO-CNN	46.30%	18,518.20	-32.18%	65	25	10,000	-45.06%	44	25	-42.33%	
300015.SZ	MMSF-ANN	48.75%	53,144.25	-5.80%	61	27	10,000	-38.89%	27	27	-44.93%	
	MMSF-BO-ANN	56.25%	60,597.77	12.83%	48	26	10,000	-9.97%	27	26	-44.06%	
	MMSF-SVM	42.08%	56,128.00	-20.75%	75	24	10,000	-48.42%	19	19	-41.77%	
	MMSF-BO-SVM	45.83%	100,422.86	11.68%	61	19	10,000	-33.77%	19	19	-44.13%	
	MMSF-CNN	38.33%	141,878.70	-4.51%	77	21	10,000	-56.20%	17	17	-43.57%	
	MMSF-BO-CNN	45.83%	93,369.53	-7.85%	100	33	10,000	-62.14%	26	25	-40.24%	

Table 10 (continued)

Stock code	Method	Accuracy	Strategy1				Strategy2				BHS	
			$v_{money}(CNY)$	Profit	N_{buy}	N_{sell}	$v_{money}(CNY)$	Profit	N_{buy}	N_{sell}	Profit	
000063.SZ	MMSF-ANN	53.33%	26,706.50	-5.79%	73	31	10,000	-17.99%	52	31	-9.69%	
	MMSF-BO-ANN	55.93%	26,863.62	-4.65%	80	35	10,000	-19.85%	55	35	-9.31%	
	MMSF-SVM	44.07%	30,236.24	-6.08%	68	24	10,000	-23.72%	42	24	-9.18%	
	MMSF-BO-SVM	47.04%	49,551.71	-2.35%	72	26	10,000	-21.58%	41	26	-8.87%	
	MMSF-CNN	45.93%	37,728.33	-13.43%	99	29	10,000	-39.21%	51	29	-9.47%	
	MMSF-BO-CNN	50.37%	24,583.75	-11.41%	83	38	10,000	-35.07%	56	38	-9.50%	
300760.SZ	MMSF-ANN	46.67%	843,801.22	-4.91%	95	34	10,000	-47.99%	37	34	-15.04%	
	MMSF-BO-ANN	51.48%	427,289.21	-7.58%	102	35	10,000	-34.53%	48	35	-15.13%	
	MMSF-SVM	43.33%	1,045,884.16	2.83%	113	22	10,000	-18.16%	37	22	-14.12%	
	MMSF-BO-SVM	47.41%	674,065.01	-12.11%	119	25	10,000	-37.57%	42	25	-14.73%	
	MMSF-CNN	39.63%	1,220,270.58	0.23%	112	23	10,000	-28.49%	34	23	-14.10%	
	MMSF-BO-CNN	42.96%	1,090,734.35	-2.79%	122	33	10,000	-49.87%	35	33	-13.78%	
002032.SZ	MMSF-ANN	50.37%	65,370.63	-15.46%	68	27	10,000	-47.68%	24	24	-23.64%	
	MMSF-BO-ANN	52.96%	72,499.84	-4.68%	75	28	10,000	-34.23%	29	28	-21.70%	
	MMSF-SVM	43.33%	78,966.45	-16.61%	66	27	10,000	-47.58%	22	22	-23.18%	
	MMSF-BO-SVM	47.04%	65,920.13	-19.80%	63	25	10,000	-48.48%	19	19	-23.11%	
	MMSF-CNN	38.89%	102,029.69	-12.54%	83	22	10,000	-51.66%	18	18	-23.58%	
	MMSF-BO-CNN	49.63%	78,943.30	-16.22%	97	33	10,000	-54.67%	23	23	-21.26%	
Average	MMSF-ANN	51.22%	139,168	-10.85%	68.1	27.6	10,000	-100.19%	35	26.6	-25.23%	
	MMSF-BO-ANN	54.70%	92,630.59	-3.52%	57.9	26.9	10,000	-16.30%	36.1	26.4	-24.09%	
	MMSF-SVM	44.88%	161,882.8	-12.08%	66.4	20.4	10,000	-33.59%	29	19.5	-23.98%	
	MMSF-BO-SVM	47.58%	122,628.1	-9.17%	62.1	20.3	10,000	-28.51%	28.4	19.2	-23.78%	
	MMSF-CNN	42.76%	184,120.3	-18.54%	74.6	22.2	10,000	-40.58%	31.1	21	-25.27%	
	MMSF-BO-CNN	46.60%	172,214.5	-16.41%	89.2	29.9	10,000	-44.83%	36.2	26.2	-24.02%	

one hand, the collected raw stock data is used to extract and classify the key points of stock trading signals. On the other hand, a more comprehensive feature factor library is generated. In the feature selection stage, three different feature selection methods are used to rank the importance of features from three different evaluation perspectives, and finally, the features with higher importance are used as input features through the cross-selection method. A Bayesian-optimized multilayer perceptron classifier is used to train and predict stock data during the model training and prediction phases. Finally, the performance of the model is evaluated from two perspectives of prediction accuracy and investment return, respectively. To illustrate the performance of the models, a series of comparative models such as IEMFCE-ANN, IEMFCE-BO-SVM, IEMFCE-SVM, IEMFCE-BO-CNN, IEMFCE-CNN, and BHS are set up in this paper for comparison experiments. The experimental results show that the proposed model is effective in uptrend, steady trend, and downtrend, and can obtain better gains.

However, there are still many issues that need to be addressed in the future. First, the optimization of the hyperparameters of the neural network is crucial for the model performance, and further research on superior optimization

algorithms is needed in the future. Second, different investment strategies have a great impact on the investment returns obtained. When accurate trading signals are obtained, choosing an irrational strategy may lead to undesirable profits. In the future, the use of artificial intelligence algorithms to simulate human behavior for stock trading can be considered.

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Data Availability The data that support the findings of this study are sourced from www.tushare.com.

Declarations

Ethical Approval This article does not contain any studies with human participants or animals performed by any of the authors.

Consent to Participate Informed consent was obtained from all individual participants included in the study.

Conflict of Interest The authors declare no competing interests.

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